



Air power or Power of Ventilation 16

Saturday

August 2008

Power of ventilation or airpower is the rate at which work is done to

maintain the air-flow through a system, and is given by the relation

$$\text{Air power} = -M \int V \cdot dp$$

where

M = mass flow rate of air through the system, kg/s

V = specific volume of air, m^3/kg .

For incompressible flow, above equation can be written as

$$\text{Air power} = MV(P_1 - P_2) = Q(P_1 - P_2)$$

Q = airflow rate = MV , m^3/s .

Neglecting change in potential and kinetic energy of air, above equation again reduces to

$$\text{Air Power} = \Delta P \cdot Q, \text{ W} = \Delta P \cdot Q \times 10^{-3} \text{ kW}$$

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where ΔP in Pascals and

Q in m^3/s

Since $\Delta P = RQ^2$

$$\text{Airpower} = RQ^3 \times 10^{-3} \text{ kW} \quad \text{or} \quad \text{Air power} \propto Q^3$$

Reqd. output of the prime mover driving a fan causing the air-flow MEMO / TO DO can be found by dividing air power by the efficiency of the fan and gear.

$$\text{Power reqd. to drive a fan} = \frac{\text{Air power}}{\eta}$$

where η = efficiency of fan and gear

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H.P. of Ventilation

The useful power developed by an air current or the H.P. in the air



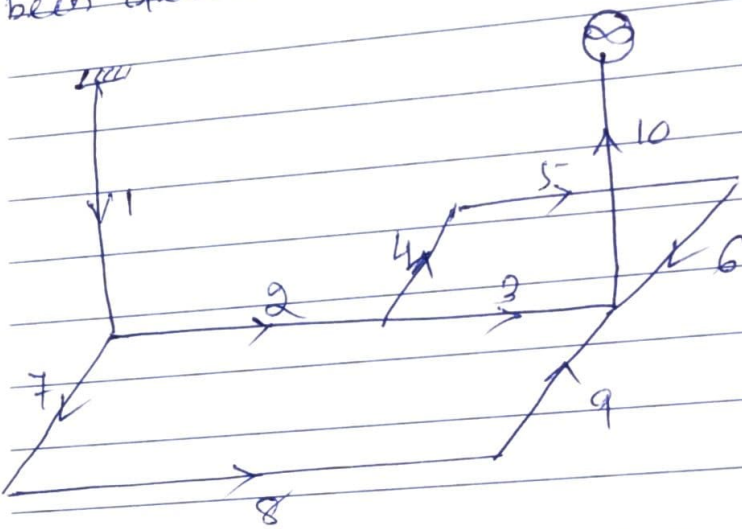
$$\text{Air H.P.} = \frac{\Delta P Q}{75} = \frac{R Q^3}{75}$$

ΔP = Ventilating pressure or total pressure, Pa

Q = Quantity in m^3/s

R = Resistance in kilomung. $\rightarrow 1 \text{ Mung} = 0.00981 \text{ NS}^2/m^8$

For the simple ventilation circuit shown in fig, the following values of resistance for the individual airways have been determined;



$$R_1 = 0.0559 \text{ NS}^2/m^8$$

$$R_2 = 0.1342 \text{ "}$$

$$R_3 = 0.1118 \text{ "}$$

$$R_4 = 0.0838 \text{ "}$$

$$R_5 = 0.1399 \text{ "}$$

$$R_6 = 0.1453 \text{ "}$$

$$R_7 = 0.1062 \text{ "}$$

$$R_8 = 0.1677 \text{ "}$$

$$R_9 = 0.1569, \text{ " } R_{10} = 0.0447$$

Determine the equivalent resistance for the entire circuit system and the total head loss of the airways, if the fan is exhausting air at the rate of $47.19 \text{ m}^3/s$. Also calculate the quantity of air flowing through each airway and total air power.

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MEMO / TO DO

Equivalent Orifice of an Airway or Mine



Murgue expressed the resistance of a mine or an airway in terms of the area of an equivalent orifice.

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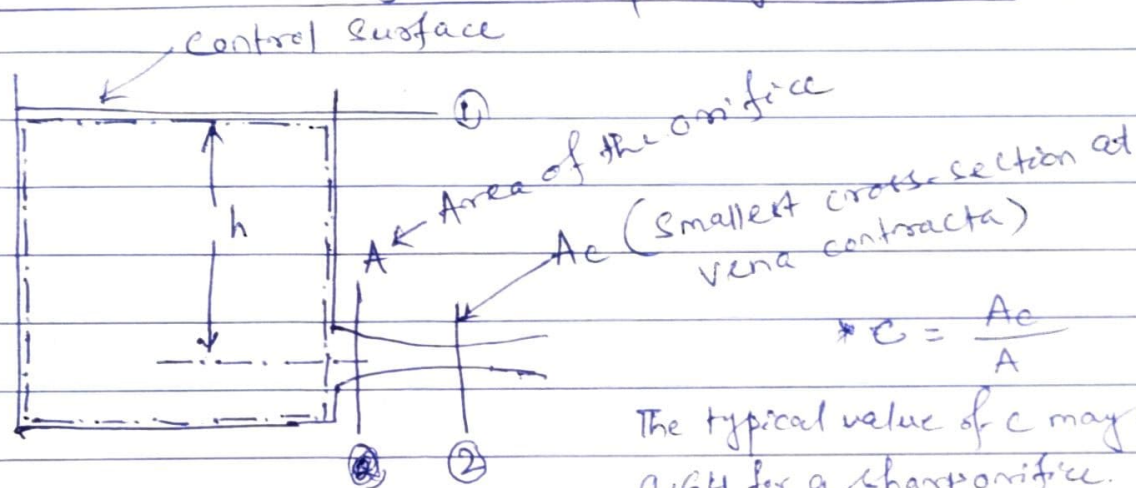
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The equivalent orifice of a mine is the area of an imaginary opening in a thin plate which offers the same resistance to the passage of air as is offered by the mine itself.

It is the area of opening through which would pass the same quantity of air as passes through the mine under the same pressure.

$$\text{Equivalent orifice} = A = \frac{1.2 Q}{\sqrt{P}} = \frac{1.2}{\sqrt{R}}, \text{ m}^2$$

Analysis of flow through a sharp-edged orifice: (Derivation)



Let Q = quantity of air flowing through the mine opening (m^3/s)

A = Area of the orifice, m^2

A_c = " of flow at the vena contracta, m^2 , $= 0.64A$

($\because$ coefficient of contraction, $c = \frac{A_c}{A} \Rightarrow A_c = c \cdot A$)

V = velocity of flow at the vena contracta

MEMO / TO DO h = head of air causing flow, m of air column

Since

Potential energy = Velocity energy

$$\text{f.g. } h = \text{f.v. } \frac{V^2}{2}$$

$$\Rightarrow V = \sqrt{2gh}$$

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$$\text{or } \frac{Q}{A_c} = \sqrt{2gh}$$



$$\text{or } A_c = \frac{Q}{\sqrt{2gh}}$$

$$\left[\frac{A_c}{A} = c = 0.64, \Rightarrow A_c = 0.64A \right]$$

$$\text{or } 0.64A = \frac{Q}{\sqrt{2gh}}$$

$$\text{or } A = \frac{Q}{0.64\sqrt{2gh}}$$

If P = Pressure causing the flow, Pa.

the $P = \rho \cdot g \cdot h$ or $\boxed{h = \frac{P}{\rho \cdot g}}$

$$\text{or } A = \frac{Q \cdot \sqrt{\rho g}}{0.64\sqrt{2P}} = \frac{Q\sqrt{\rho}}{0.64\sqrt{2P}}$$

Assuming standard value of $\rho = 1.2 \text{ kg/m}^3$

$$\boxed{A = 1.2 \cdot \frac{Q}{\sqrt{P}} = \frac{1.2Q}{\sqrt{P}}}, \text{ m}^2$$

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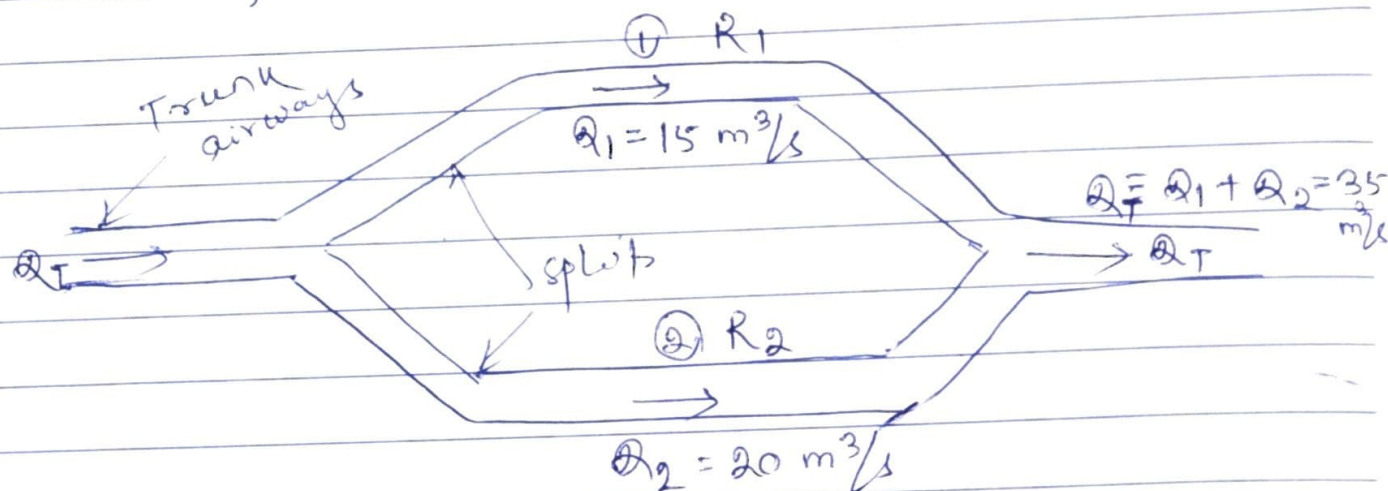
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MEMO / TO DO

Example 22 : Two ventilation splits 1 and 2 pass 15 and 20 m³/s of air respectively with a pressure drop of 500 Pa across them. The trunk airways consume a pressure of 300 Pa. If the air flow in the two splits is to be equal by installing a regulator, calculate the size of the regulator, assuming the fan pressure to remain constant before and after installation of a regulator. What will be the air flow in the mine after fitting the regulator.



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In order to make the air flow in the two splits equal, a regulator has to be installed in split 2.

Let the resistance of the regulator be R_R and the new quantity of air flowing in the mine after installation of regulator be Q . The condition for the air flow in both the splits to be equal is

$$R_1 \left(\frac{Q}{2} \right)^2 = (R_2 + R_R) \left(\frac{Q}{2} \right)^2$$

or $R_1 = R_2 + R_R$

MEMO / TO DO : i.e. the resistance of split 1 has be equal with the ^{combined} resistance of the split 2 and the installed regulator

or $R_R = R_1 - R_2$

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where

$$R_1 = \text{Resistance of split ①} = \frac{500}{15^2} = 2.22 \text{ N s}^2/\text{m}^8$$

$$R_2 = \text{Resistance of split ②} = \frac{500}{20^2} = 1.25 \text{ N s}^2/\text{m}^8$$

if R_T = resistance of the trunk airways,

$$R_T = \frac{300}{35^2} = 0.245 \text{ N s}^2/\text{m}^8$$

$$\therefore R_R = R_1 - R_2 = 2.22 - 1.25 = 0.97 \text{ N s}^2/\text{m}^8$$

$$\therefore \text{Area of the regulator} = \frac{1.2}{\sqrt{R_R}} = \frac{1.2}{0.97} = 1.22 \text{ m}^2$$

Calculation of air flow in the mine after installation of regulator

Total pressure loss in the mine = Pressure drop across the trunk airways + pressure drop across the splits
 $= 300 + 500 = 800 \text{ Pa}$

$$\text{or } 800 = R_T \cdot Q^2 + R_1 \cdot \left(\frac{Q}{2}\right)^2 = 0.245 Q^2 + 2.22 \cdot \frac{Q^2}{4}$$

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$$\text{or } Q^2 = \frac{800}{0.8} = 1000$$

$$\text{or } Q = \sqrt{1000} = 31.6 \text{ m}^3/\text{s}$$

(Ans 1)

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